INCH-POUND
MIL-DTL-16377/9B(SH)
30 June 2014
SUPERSEDING
MIL-F-16377/9A(SH)
29 July 1981

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, GENERAL LIGHTING, 24 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 73.3, 73.4, 346.2, 346.3, 73.3SSL, 73.4SSL, 346.2SSL, AND 346.3SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

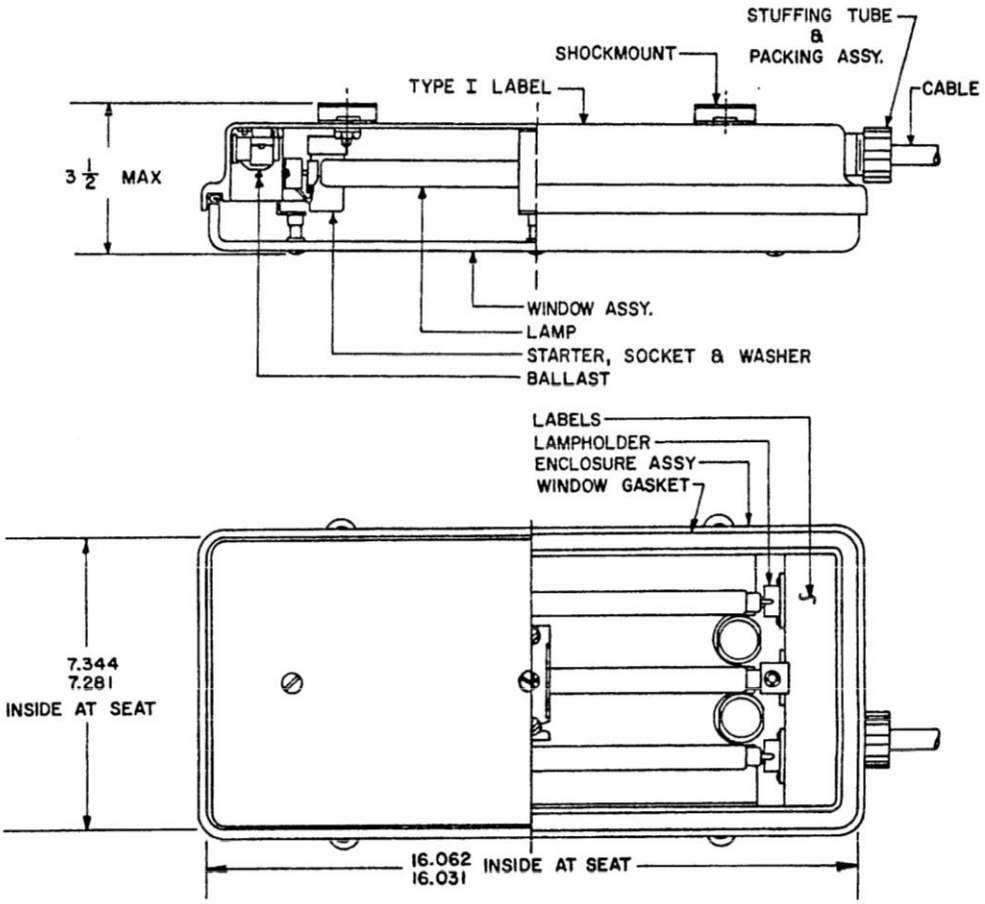


FIGURE 1. Typical view of symbols 73.3 and 346.2.

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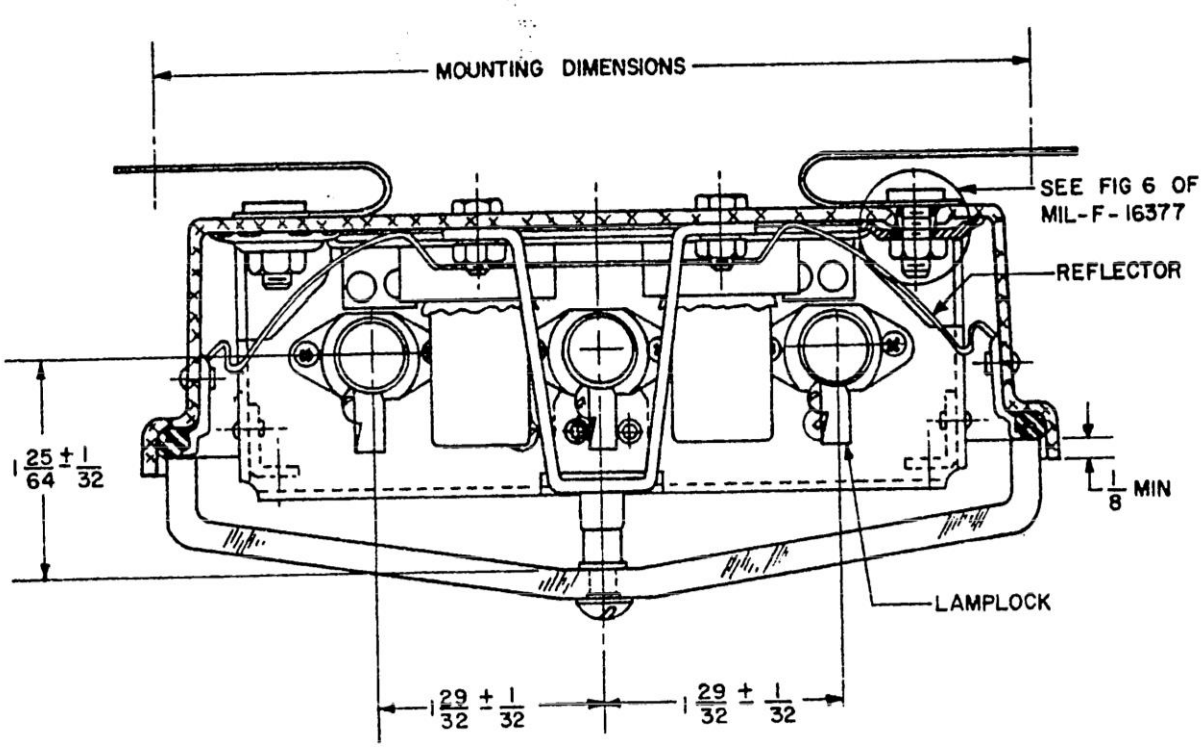


FIGURE 2. Typical section for symbols 73.3 and 346.2.

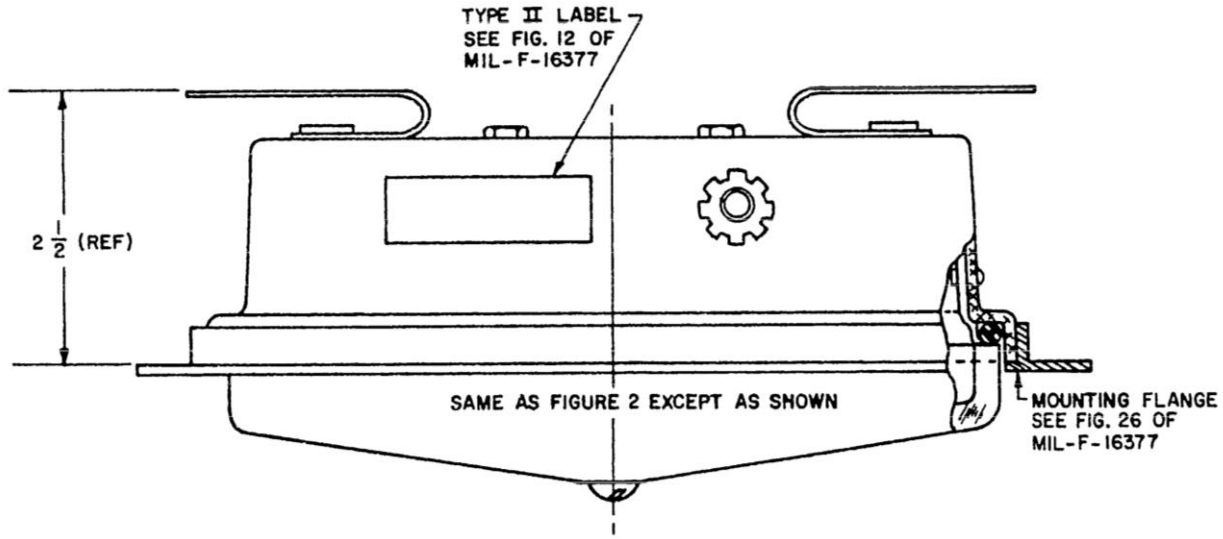
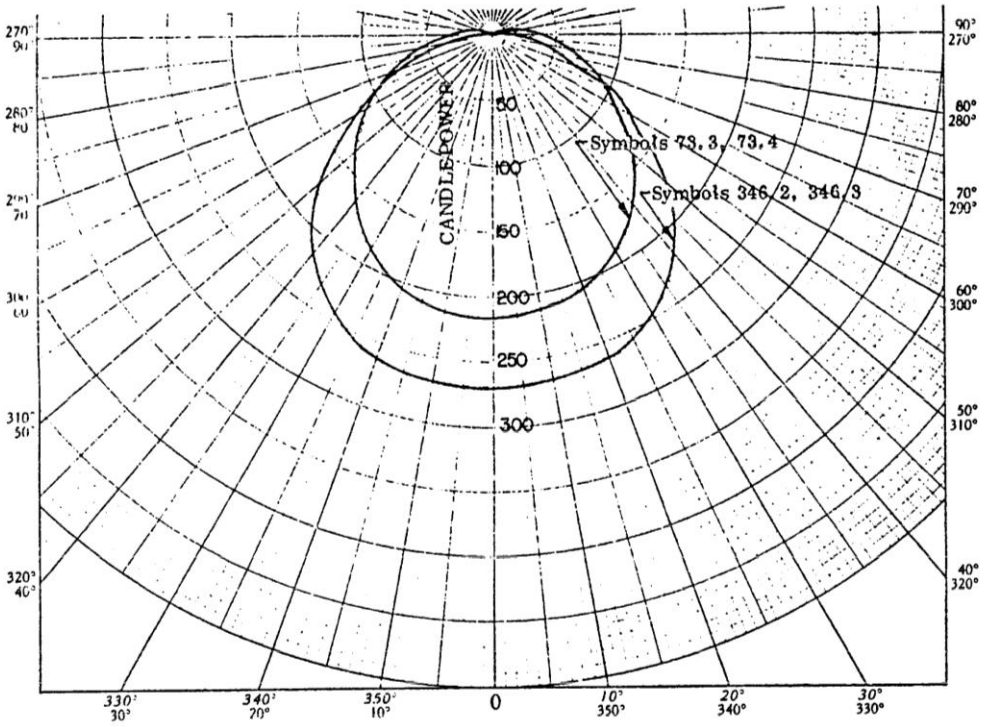


FIGURE 3. Typical section for symbols 73.4 and 346.3.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	73.3, 73.4	346.2, 346.3
0	212	265
5	212	265
10	210	265
15	208	260
20	202	260
25	195	251
30	190	242
35	180	232
40	165	218
45	155	200
50	138	175
55	122	147
60	112	125

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	73.3, 73.4	346.2, 346.3
65	95	97
70	85	75
75	70	57
80	55	40
85	45	30
90	32	20
95	22	15
100	15	10
105	10	11
110	5	6
115	5	0
120	5	0
125	0	0

PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Plane:	At right angles to long axis of fixture through geometric center of lamp.
Voltage:	115 volts, 60 hertz.
Ambient temperature:	76 °F (24.4 °C).
Lamps:	Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast:	Use ballast furnished with fixture.
Lamp correction:	Distribution curves based on 380 lumens per lamp.

FIGURE 4. Candlepower distribution curves.

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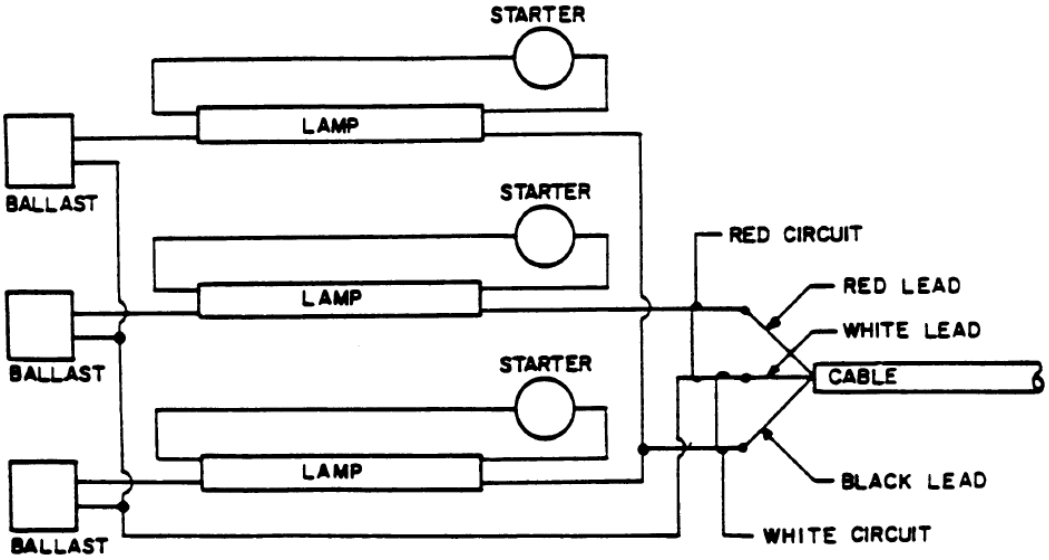


FIGURE 5. Wiring diagram.

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBER:

73.3: Complete lighting fixture assembly, standard mounting, with three 8-watt cool white fluorescent lamps and white diffusing window assembly.

73.4: Complete lighting fixture assembly, flush mounting, with three 8-watt cool white fluorescent lamps and white diffusing window assembly.

346.2: Complete lighting fixture assembly, standard mounting, with three 8-watt cool white fluorescent lamps and clear and white prismatic window assembly.

346.3: Complete lighting fixture assembly, flush mounting, with three 8-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:

Electrical rating: 24 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration: See figures 1, 2, and 3 and figure 10 of MIL-DTL-16377.
Photometric data: See figure 4.
Lampholder spacing: See figure 21 of MIL-DTL-16377.
Wiring diagram: See figure 24 of MIL-DTL-16377.
Mounting dimensions:
Symbols 73.3 and 346.2: See figure 25 of MIL-DTL-16377.
Symbols 73.4 and 346.3: See figure 26 of MIL-DTL-16377.
Window assembly:
Symbols 73.3 and 73.4: Military part number M16377/9-001, figure 27 of MIL-DTL-16377.
Symbols 346.2 and 346.3: Military part number M16377/9-003, figure 27 of MIL-DTL-16377.
Enclosure: Watertight. See 3.5.11 of MIL-DTL-16377.
Cable: 10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:
Symbols 73.3 and 346.2: 7 pounds 7 ounces maximum.
Symbols 73.4 and 346.3: 8 pounds 11 ounces maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbols 73.4 and 346.3 only.
c. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table I and window assemblies of table V by name and stock number only.

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

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SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBER:

73.3SSL: Complete lighting fixture assembly, standard mounting, with LED luminaire and white diffusing window assembly.
73.4SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and white diffusing window assembly.
346.2SSL: Complete lighting fixture assembly, standard mounting, LED luminaire and clear and white prismatic window assembly.
346.3SSL: Complete lighting fixture assembly, flush mounting, with LED luminaire and clear and white prismatic window assembly.

DESIGN:

Electrical rating:	24 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1, 2, 3 and 4 and figure 10 of MIL-DTL-16377.
Photometric data:	See figure 4 and 3.7.1 of MIL-DTL-16377.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Shall maintain same control scheme for light output as legacy fluorescent fixture. Refer to figure 24 of MIL-DTL-16377.
Mounting dimensions:	
Symbols 73.3SSL and 346.2SSL:	See figure 25 of MIL-DTL-16377.
Symbols 73.4SSL and 346.3SSL:	See figure 26 of MIL-DTL-16377.
Window assembly:	
Symbols 73.3SSL and 346.2SSL:	Military part number M16377/7-001, figure 27 of MIL-DTL-16377.
Symbols 73.4SSL and 346.3SSL:	Military part number M16377/7-002, figure 27 of MIL-DTL-16377.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Cable:	10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Weight:	
Symbols 73.3SSL and 346.2SSL:	7 pounds 7 ounces maximum.
Symbols 73.4SSL and 346.3SSL:	8 pounds 11 ounces maximum.
Designation and marking:	The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbols 70.4SSL and 336.1SSL only.
c. One Type IV. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:
   (1) A list of all standard parts shown in table I and window assemblies of table V by name and stock number only, if appropriate, or
   (2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

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EXAMINATION AND TESTS:

Examinations and tests shall be as specified in MIL-DTL-16377 and the following:

First article inspection:
Complete fixtures: See table II for Type I and table III for Type III.
Window assemblies: See table IV.
Conformance:
Comparison inspection: Same as first article inspection.
Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table V.

TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	ANSI/ANSLG C78.81	6240-00-299-5546	Industry number F8T5/CW
Lamp lock	M16377/40-001	6250-00-399-7039	
Lampholder	UL 496	6250-00-377-8590	SUPP. SA, FIG. SA2.1
Startersocket	UL 496	6250-00-295-2738	SUPP. SA, FIG. SA2.4
Starter	NEMA ANSI C78.180	6250-00-299-5962	Commercial designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Shock mount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished, see 3.7.9 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-002	6210-00-503-2810	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-014	6210-01-313-8170	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-013	6210-01-351-3943	Not furnished, see 3.7.9 of MIL-DTL-16377

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TABLE II. First article and comparison inspection for complete type I fixtures.[1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE III. First article and comparison inspection for complete type III fixtures, symbols 73.3SSL, 73.4SSL, 346.2SSL and 346.3SSL. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight, unless otherwise specified.
Shock	3.5.12	4.8.7	[2/]
Vibration	3.5.13	4.8.8	[2/]
Enclosure effectiveness	3.5.11	4.8.14.4	[2/] Watertight, unless otherwise specified.
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Noise	3.5.14	4.8.9	[2/]
Continuity of grounding	3.6.13.4	4.8.12	[2/]
Photometrics	3.7.1	4.8.6	
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT), white light only	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

[2/] In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE IV. First article and comparison inspection for window assemblies. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.6.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

NOTE:
[1/] Requirement and test method paragraph numbers refer to MIL-DTL-16377.

TABLE V. Part number and general information.

Military Part [1/] Number M16377/9-	Item Name	National Stock Number (NSN)	Remarks
73.3	Complete fixture, Symbol 73.3	6210-00-295-2880	[2/]
73.4	Complete fixture, Symbol 73.4	6210-01-088-2512	[2/]
346.2	Complete fixture, Symbol 346.2	6210-01-088-6944	
346.3	Complete fixture, Symbol 346.3	6210-01-186-9957	
73.3SSL	Complete fixture, solid state, Symbol 73.3SSL		
73.4SSL	Complete fixture, solid state, Symbol 73.4SSL		
346.2SSL	Complete fixture, solid state, Symbol 346.2SSL		
346.3SSL	Complete fixture, solid state, Symbol 346.3SSL		
001	Window, white diffusing	6210-00-548-0214	[3/]
003	Window, clear and white prismatic	6210-01-312-0377	

NOTES:
[1/] Military part numbers M16377/9-346, 346.1, and 002 have been superseded by military part numbers M16377/9-346.2, 346.3, and M16377/67-002, respectively.
[2/] Formerly covered on NAVSHIPS Drawing 9000-S6401-74461 and MS17150.
[3/] Formerly covered on MS1689.38.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
(Project 6210-2012-024)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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